

Communication-Enhanced MAPPO for Coordinated ACP Teams in Strike-EA Missions

Mid-Term Report

by

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This thesis is confidential and cannot be made public.

Cover: Artist impression of autonomous collaborative platforms (ACPs).
Adapted from [1].

Methodology

This chapter presents the cooperative striker-jammer problem, the two simulation environments used in this work, the baseline flattened MAPPO algorithm, and the proposed FOFE-MAPPO variant.

1.1. Problem Formulation

The cooperative strike-EA engagement is modelled as a decentralised partially-observable Markov decision process (Dec-POMDP) $\langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}, \{\mathcal{O}_i\}, P, \mathcal{R}, \gamma \rangle$ on a bounded two-dimensional plane $\mathcal{W} = [0, 1]^2$ with reflecting boundaries. The agent set is partitioned by role into strikers and jammers,

$$\mathcal{N} = \mathcal{N}_s \cup \mathcal{N}_j, \quad |\mathcal{N}_s| = N_s, \quad |\mathcal{N}_j| = N_j, \quad (1.1)$$

operating in an environment that additionally contains a set of radars $\mathcal{R} = \{1, \dots, N_r\}$ and protected targets $\mathcal{T} = \{1, \dots, N_t\}$. The team objective is to destroy $\mathcal{T}$; jammers degrade radar coverage so that strikers may penetrate the defended airspace. Training follows the centralised-training / decentralised-execution (CTDE) paradigm: during execution each agent conditions only on its local observation o_i , while during training the critics have access to the global state $s \in \mathcal{S}$.

1.2. Agent Kinematics

Each agent $i \in \mathcal{N}$ follows a planar unicycle model with double-integrator heading dynamics. The continuous agent state is $\mathbf{s}_i = [x_i, y_i, \psi_i, v_i, \dot{\psi}_i]^\top$, giving position, heading, linear speed and heading rate. One environment step integrates the commanded acceleration $a_i^v \in [-1, 1]$ and angular-acceleration multiplier $a_i^\omega \in [-1, 1]$ through

$$v_{i,t+1} = \text{clip}(v_{i,t} + a_i^v \Delta v_{\max}, v_{\min}, v_{\max}), \quad (1.2)$$

$$\dot{\psi}_{i,t+1} = \text{clip}(\dot{\psi}_{i,t} + a_i^\omega \Delta \dot{\psi}_{\max}, -\dot{\psi}_{\max}^*(v), \dot{\psi}_{\max}^*(v)), \quad (1.3)$$

with a speed-dependent turn-rate limit that enforces a minimum turn radius $R_{\min}$,

$$\dot{\psi}_{\max}^*(v) = \min(\dot{\psi}_{\max}, v/R_{\min}). \quad (1.4)$$

Heading and position are then propagated by explicit Euler integration with step size Δt ,

$$\psi_{i,t+1} = (\psi_{i,t} + \dot{\psi}_{i,t+1} \Delta t) \bmod 2\pi, \quad \begin{bmatrix} x_{i,t+1} \\ y_{i,t+1} \end{bmatrix} = \begin{bmatrix} x_{i,t} \\ y_{i,t} \end{bmatrix} + v_{i,t+1} \Delta t \begin{bmatrix} \cos \psi_{i,t+1} \\ \sin \psi_{i,t+1} \end{bmatrix}. \quad (1.5)$$

Positions are clipped to $\mathcal{W}$. The action head is factorised, with a_i^v and a_i^ω each drawn from an independent categorical distribution over $K = 7$ bins mapping to multipliers $c_k \in \{-1, -0.5, -0.1, 0, +0.1, +0.5, +1\}$. Parameters are shared within each role. Numerical values are listed in Table 1.1.

Table 1.1: Agent kinematic parameters (normalised units: 1 world unit = 1000 km, $\Delta t = 60$ s).

Symbol	Meaning	Value
$v_{\max}$	Maximum speed	0.020
$v_{\min}$	Minimum speed (per role)	0.005
$\Delta v_{\max}$	Linear-acceleration step	0.010
$\dot{\psi}_{\max}$	Maximum heading rate	$12^\circ/\text{step}$
$\Delta \dot{\psi}_{\max}$	Angular-acceleration step ($=0.1 \dot{\psi}_{\max}$)	$1.2^\circ/\text{step}$
$R_{\min}$	Minimum turn radius	0.001
K	Discrete bins per action dimension	7

1.3. Observation, Communication and Action Spaces

Sensing. Each agent has two range parameters: a sensing radius R_{obs} within which it directly detects other entities, and a communication radius R_{comm} defining direct peer links. The direct neighbourhood and the multi-hop communication subgroup of agent i are

$$\mathcal{N}_i^1 = \{i' \in \mathcal{N} \setminus \{i\} \mid \rho_{ii'} \leq R_{\text{comm}}\}, \quad \mathcal{S}_i = \{i\} \cup \text{TC}(\mathcal{N}_i^1), \quad (1.6)$$

where $\rho_{ab} = \|\mathbf{p}_a - \mathbf{p}_b\|_2$ and $\text{TC}(\cdot)$ is the transitive closure over the graph induced by $\mathcal{N}^1$. Members of $\mathcal{S}_i$ pool their local detections, so the set of entities of type $\xi \in \{\text{ag}, \text{tg}, \text{rd}\}$ available to agent i is

$$\mathcal{D}_i^\xi = \mathcal{K}^\xi \cup \bigcup_{i' \in \mathcal{S}_i} \left\{ e \in \xi \mid \rho_{i'e} \leq R_{\text{obs}} \right\}, \quad (1.7)$$

where $\mathcal{K}^\xi \subseteq \xi$ denotes the subset of entities assumed *known a priori* (e.g. briefed targets/radars); all other entities must be detected through (1.7). The same multi-hop sharing logic is used for the baseline and the FOFE encoder; the two methods differ only in how $\mathcal{D}_i^\xi$ is embedded into the policy input.

Observation vector. Each agent's local observation decomposes into a fixed-size ego vector and three typed entity channels,

$$o_i = (o_i^{\text{self}}, \{o_{ii'}^{\text{ag}}\}_{i' \in \mathcal{D}_i^{\text{ag}}}, \{o_{ij}^{\text{tg}}\}_{j \in \mathcal{D}_i^{\text{tg}}}, \{o_{ik}^{\text{rd}}\}_{k \in \mathcal{D}_i^{\text{rd}}}). \quad (1.8)$$

The ego vector uses normalised kinematics and an alive flag ($d_{\text{self}} = 6$). Each entity feature contains relative range and bearing expressed in the body frame of agent i plus type-specific flags ($d_{\text{ag}} = 4$, $d_{\text{tg}} = 3$, $d_{\text{rd}} = 3$).

Strike and jam effects. A striker $i \in \mathcal{N}_s$ has a forward engagement cone with range r^{Stk} and half-angle $\frac{1}{2}\alpha^{\text{Stk}}$; the engaged target set is

$$\mathcal{A}_i^{\text{Stk}} = \{j \in \mathcal{T} \mid \rho_{ij} \leq r^{\text{Stk}} \wedge |\theta_{ij}| \leq \frac{1}{2}\alpha^{\text{Stk}}\}, \quad (1.9)$$

with θ_{ij} the body-frame bearing of j . Any $j \in \mathcal{A}_i^{\text{Stk}}$ is removed from $\mathcal{T}$ deterministically. The jamming model is environment-specific and is defined in Section 1.4.

1.4. Environment Models

Two environments are used. Both share the kinematics, observation model, communication graph and strike mechanics described above; they differ only in how each radar's effective detection range is computed.

1.4.1. Low-Fidelity Environment

The low-fidelity (LF) environment models radar effectiveness by a single binary jam state. The effective detection range of radar $k \in \mathcal{R}$ is

$$R_k^{\text{eff}} = \begin{cases} \max(0, R_k - \Delta R), & \text{if } |\mathcal{N}_j^{\text{alive}}| \geq 1, \\ R_k, & \text{otherwise,} \end{cases} \quad (1.10)$$

with R_k the nominal range and ΔR a fixed range reduction. An agent i is killed in a given step if $\rho_{ik} \leq R_k^{\text{eff}}$ for any $k \in \mathcal{R}$, with a per-step kill probability p_{kill} . Equation (1.10) assumes isotropic jamming, i.e. a single alive jammer suffices to suppress all radars regardless of relative geometry. Low-fidelity parameters are listed in Table 1.2.

Table 1.2: Low-fidelity radar / jam parameters.

Symbol	Meaning	Value
R_k	Nominal radar detection range	0.20
ΔR	Jammer-induced range reduction	0.15
p_{kill}	Per-step kill probability inside R_k^{eff}	1.0
r^{Stk}	Striker engagement range	0.10
α^{Stk}	Striker engagement full-cone angle	60°

1.4.2. High-Fidelity Environment

The high-fidelity (HF) environment replaces the binary jam model with an angular burn-through model based on the monostatic radar equation. The LF kinematics, communication and reward structure are *unchanged*; only the mapping from radar/jammer geometry to R_k^{eff} is replaced. Three effective ranges are defined per radar: the unconstrained range R_{unc} , the main-lobe burn-through range R_{main} , and the side-lobe burn-through range R_{side} .

The unconstrained range follows from equating the received target echo power to the receiver noise floor at a prescribed minimum detection SNR Γ_{min} ,

$$R_{\text{unc}} = \left(\frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 k T_0 B_n L \Gamma_{\text{min}}} \right)^{1/4}. \quad (1.11)$$

Jamming is modelled via the burn-through condition, where the received jammer power equals the target echo at R_k^{eff} . For main-lobe jamming (the jammer lies inside the radar main-lobe sector) the effective range reduces to

$$R_{\text{main}} = \left(\frac{P_t G_t \sigma}{4\pi P_J G_J} R_J^2 \right)^{1/4}, \quad (1.12)$$

and for side-lobe jamming, accounting for the reduced receive gain G_S toward the jammer,

$$R_{\text{side}} = \left(\frac{\sigma}{4\pi} \cdot \frac{P_t G_t}{P_J G_J} \cdot \frac{G_t R_J^2}{G_S} \right)^{1/4}. \quad (1.13)$$

Here P_t, P_J denote radar and jammer transmit powers, G_t, G_r, G_J, G_S the corresponding antenna gains, λ the wavelength, σ the target radar cross section, R_J the radar-jammer range, k the Boltzmann constant, T_0 the system temperature, B_n the receiver bandwidth and L the system losses. For a monostatic radar we use $G_r = G_t$.

Let θ_{kJ} denote the bearing of jammer J relative to radar k , and let $\theta_{\text{main}}, \theta_{\text{side}}$ be the configured main- and side-lobe half-widths. For any agent i at bearing θ_{ki} from radar k , the per-agent, per-radar effective range is a piecewise function of which jammer sector the agent lies in,

$$R_{ki}^{\text{eff}} = \begin{cases} \min_{J \in \mathcal{N}_j^{\text{alive}}} R_{\text{main}}(R_{kJ}), & |\theta_{ki} - \theta_{kJ}| \leq \theta_{\text{main}} \text{ for some } J, \\ \min_{J \in \mathcal{N}_j^{\text{alive}}} R_{\text{side}}(R_{kJ}), & \theta_{\text{main}} < |\theta_{ki} - \theta_{kJ}| \leq \theta_{\text{side}} \text{ for some } J, \\ R_{\text{unc}}, & \text{otherwise.} \end{cases} \quad (1.14)$$

Kill mechanics are identical to the LF case but with the per-agent range (1.14). Parameters are summarised in Table 1.3.

Table 1.3: High-fidelity radar / jammer RF parameters. Gains and losses are specified in dB and converted to linear scale internally. Ranges are computed in SI units and converted to world units via a scaling factor $s_{m \rightarrow w}$.

Symbol	Meaning	Value
P_t	Radar transmit power	10^6 W
G_t	Radar transmit / receive gain	40 dB
λ	Wavelength	0.03 m
σ	Target radar cross section	1 m^2
T_0	System temperature	290 K
B_n	Receiver bandwidth	10^6 Hz
L	System losses	4 dB
$\Gamma_{\min}$	Minimum detection SNR	0 dB
G_S	Radar side-lobe gain	5 dB
P_J	Jammer transmit power	10 W
G_J	Jammer antenna gain	15 dB
θ_{main}	Main-lobe half-width	3°
θ_{side}	Side-lobe half-width	5°
$s_{m \rightarrow w}$	Metres per world unit	10^6

Figure placeholder: a polar diagram of the HF radar coverage (unconstrained range, main-lobe notch, side-lobe sector) for a single radar with a single jammer would help the reader visualise Eq. (1.14).

1.5. Reward Structure

Rewards are emitted per alive agent at every step and consist of sparse event terms and dense shaping terms,

$$r_{i,t} = r_{i,t}^{\text{sparse}} + r_{i,t}^{\text{dense}}, \quad r_{i,t}^{\text{dense}} = r^{\text{time}} + r_{i,t}^{\text{border}} + r_{i,t}^{\text{radar}} + r_{i,t}^{\text{app}} + r_{i,t}^{\text{form}} + r_{i,t}^{\text{ctrl}}. \quad (1.15)$$

Sparse terms. Target kills, agent losses and mission completion use a convex combination of a team-mean component and an individual credit,

$$r_i^e = \eta \bar{R}_e + (1 - \eta) \tilde{R}_i^e, \quad e \in \{\text{kill, loss, term}\}, \quad (1.16)$$

with $\bar{R}_e$ the per-event reward averaged over alive agents and $\tilde{R}_i^e$ the credit assigned to the individual responsible (the striker whose cone (1.9) destroyed the target, or the destroyed agent itself). The scalar $\eta \in [0, 1]$ controls the team– individual mix.

Dense terms. All dense terms share a common piecewise linear-exponential shape $\phi_\xi(d; \theta_\xi)$ parametrised by a weight pair $(w^{\text{lin}}, w^{\text{exp}})$, a knee d^{knee} , an exponent α and a cut-off d^{max} ,

$$\phi(d; \theta) = \begin{cases} 0, & d \geq d^{\text{max}}, \\ w^{\text{lin}} \frac{d^{\text{max}} - d}{d^{\text{max}} - d^{\text{knee}}}, & d^{\text{knee}} \leq d < d^{\text{max}}, \\ w^{\text{lin}} + w^{\text{exp}} \left(e^{\alpha(1-d/d^{\text{knee}})} - 1 \right), & d < d^{\text{knee}}. \end{cases} \quad (1.17)$$

Approach-type terms (striker→target, jammer→radar, formation) are expressed as distance penalties and are zero at contact; avoidance-type terms (border, radar-zone) are negative and increase as the agent

enters the hazard,

$$r_{i,t}^{\text{time}} = -c_{\text{time}}, \quad (1.18)$$

$$r_{i,t}^{\text{border}} = -\phi(d_{i,t}^{\text{edge}}; \theta_{\text{bd}}), \quad (1.19)$$

$$r_{i,t}^{\text{radar}} = -\sum_{k \in \mathcal{R}} \phi(d_{ik,t}^{\text{rd}}; \theta_{\text{rd}}), \quad (1.20)$$

$$r_{i,t}^{\text{app}} = -\phi(d_{i,t}^{\text{tg}}; \theta_{\text{app}}), \quad i \in \mathcal{N}_{\text{s}}, \quad (1.21)$$

$$r_{i,t}^{\text{form}} = -\phi(d_{i,t}^{\text{ally}}; \theta_{\text{form}}), \quad i \in \mathcal{N}_{\text{j}}, \quad (1.22)$$

$$r_{i,t}^{\text{ctrl}} = -\lambda_a (a_i^v)^2 - \lambda_\omega (a_i^\omega)^2. \quad (1.23)$$

r^{time} discourages long trajectories; r^{border} penalises proximity to the map boundary; r^{radar} penalises proximity to radar threat zones; r^{app} pulls each striker toward the nearest alive target; r^{form} keeps each jammer close to a striker; r^{ctrl} penalises aggressive control inputs. Reward parameters are listed in Table 1.4.

Table 1.4: Reward parameters. Inactive components set to zero.

Component	Parameter(s)	Value
Target kill	R_{kill}	+3
Agent loss	R_{loss}	-3
Mission complete	R_{term}	+1
Team sharing	η	0.5
Time	c_{time}	0.01
Border	$w_{\text{bd}}^{\text{lin}}, d_{\text{bd}}^{\text{max}}$	0.05, 0.05
Radar zone	$w_{\text{rd}}^{\text{lin}}, d_{\text{rd}}^{\text{max}}$	0.05, 0.10
Striker approach	$w_{\text{app}}^{\text{lin}}, d_{\text{app}}^{\text{max}}$	0.10, 1.00
Jammer formation	$w_{\text{form}}^{\text{lin}}, d_{\text{form}}^{\text{max}}$	0.10, 0.50
Control effort	$\lambda_a, \lambda_\omega$	0.01, 0.01

1.6. MAPPO Training

Both the baseline and the proposed method are trained with role-factorised MAPPO [CITE: Yu et al., MAPPO, 2022] built on the clipped PPO objective [CITE: Schulman et al., PPO, 2017]. Each role $\rho \in \{\text{s}, \text{j}\}$ has its own actor π_{θ_ρ} and centralised critic V_{ϕ_ρ} ; parameters are shared across all agents of the same role.

Advantage estimation. Rollouts of length T are collected from B parallel environments. For each role, per-step TD residuals and generalised-advantage estimates [CITE: Schulman et al., GAE, 2016] are

$$\delta_t^\rho = r_t^\rho + \gamma V_{\phi_\rho}(s_{t+1})(1 - d_t) - V_{\phi_\rho}(s_t), \quad \hat{A}_t^\rho = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l \delta_{t+l}^\rho, \quad (1.24)$$

with GAE return $\hat{R}_t^\rho = \hat{A}_t^\rho + V_{\phi_\rho}(s_t)$. Within each minibatch, advantages are standardised: $\tilde{A} = (\hat{A} - \mu_{\hat{A}})/\sigma_{\hat{A}}$.

Clipped policy loss. With importance ratio $\varrho_t(\theta) = \exp(\log \pi_{\theta_\rho}(a_t | o_t) - \log \pi_{\theta_{\rho,\text{old}}}(a_t | o_t))$,

$$\mathcal{L}_\rho^{\text{PPO}}(\theta) = -\mathbb{E}_t \left[\min(\varrho_t \tilde{A}_t^\rho, \text{clip}(\varrho_t, 1 - \epsilon, 1 + \epsilon) \tilde{A}_t^\rho) \right] - \beta \mathbb{E}_t[\mathcal{H}(\pi_{\theta_\rho})], \quad (1.25)$$

and the critic loss is the unclipped mean-squared error

$$\mathcal{L}_\rho^V(\phi) = \mathbb{E}_t \left[(V_{\phi_\rho}(s_t) - \hat{R}_t^\rho)^2 \right]. \quad (1.26)$$

Rewards are rescaled by a running second moment (standard deviation only, no mean subtraction) updated with a Welford estimator, preserving the sign of sparse terminal events. Gradients are clipped per role at $\|g\|_2 \leq g_{\max}$, and actor/critic are updated with independent Adam optimisers.

MultiCategorical distribution. The factorised action distribution is the product of two independent categoricals over the $K = 7$ bins of a_i^v and a_i^w . Logits are sanitised by $\text{clip}(\ell, -\ell_{\max}, \ell_{\max})$ with $\ell_{\max} = 20$ to guard against occasional extreme values during early exploration.

Table 1.5: MAPPO and optimiser hyperparameters (shared across baseline and FOFE-MAPPO unless stated otherwise).

Symbol	Meaning	Value
B	Parallel environments	512
T	Rollout length per environment	150
N_{it}	Training iterations	100
E	Epochs per iteration	10
M	Mini-batch size	2048
γ	Discount factor	0.99
λ	GAE trace decay	0.95
ϵ	PPO clip range	0.2
β	Entropy coefficient	0.01
$g_{\max}$	Max gradient norm	1.0
α_{π}	Actor learning rate	3×10^{-4}
α_V	Critic learning rate	3×10^{-4}

1.7. Baseline: Flattened MAPPO

The baseline uses the same MAPPO objective, communication graph (1.6)–(1.7), environment and reward structure as the proposed method, and differs only in how the variable-size entity sets $\mathcal{D}_i^{\xi}$ in (1.8) are encoded into a fixed-dimensional input.

Observation flattening. For each entity type ξ a fixed capacity K_{ξ} is chosen. The entities in $\mathcal{D}_i^{\xi}$ are sorted by range to agent i , truncated to the first K_{ξ} , and padded with zeros when $|\mathcal{D}_i^{\xi}| < K_{\xi}$. A per-slot visibility mask $m_i^{\xi} \in \{0, 1\}^{K_{\xi}}$ is concatenated with the feature block. The resulting fixed-length observation

$$o_i^{\text{flat}} = \left[o_i^{\text{self}} \parallel (o_{ii'}^{\text{ag}}, m_{ii'}^{\text{ag}})_{i'=1}^{K_{\text{ag}}} \parallel (o_{ij}^{\text{tg}}, m_{ij}^{\text{tg}})_{j=1}^{K_{\text{tg}}} \parallel (o_{ik}^{\text{rd}}, m_{ik}^{\text{rd}})_{k=1}^{K_{\text{rd}}} \right], \quad (1.27)$$

has fixed dimension $d_{\text{in}} = d_{\text{self}} + \sum_{\xi} (d_{\xi} + 1)K_{\xi}$, so the policy cannot generalise to observation sizes exceeding the training capacity. The actor and centralised critic are realised as multi-layer perceptrons with ReLU activations, depth L and width h , parameter-shared within each role (Table 1.6).

Table 1.6: Baseline flattened-MAPPO network hyperparameters.

Symbol	Meaning	Value
K_{ag}	Agent slots	$N_s + N_j - 1$
K_{tg}	Target slots	N_t
K_{rd}	Radar slots	N_r
h	MLP hidden width	256
L	MLP depth (hidden layers)	3
ϕ	Activation	ReLU

Algorithm 1 Flattened MAPPO (one training iteration).

Require: actors π_{θ_ρ} , critics V_{ϕ_ρ} , optimisers $\text{Adam}_\rho^\pi, \text{Adam}_\rho^V$; hyperparameters $(B, T, E, M, \gamma, \lambda, \epsilon, \beta, g_{\max})$.

- 1: **for** $b = 1, \dots, B$ in parallel **do**
- 2: Reset environment b ; set $\mathbf{s}_0^{(b)} \sim p_0$.
- 3: **for** $t = 0, \dots, T - 1$ **do**
- 4: Compute $\mathcal{D}_i^\xi$ via (1.6)–(1.7); build o_i^{flat} via (1.27) for each i .
- 5: Sample $a_{i,t} \sim \pi_{\theta_\rho}(\cdot \mid o_{i,t}^{\text{flat}})$.
- 6: Step env; log $(s_t, o_{i,t}^{\text{flat}}, a_{i,t}, r_{i,t}, d_t, \log \pi_{\theta_\rho^{\text{old}}})$.
- 7: **end for**
- 8: **end for**
- 9: For each role ρ , compute $V_{\phi_\rho}(s_t)$ and GAE $\hat{A}_t^\rho, \hat{R}_t^\rho$ via (1.24).
- 10: **for** $e = 1, \dots, E$ **do**
- 11: **for** minibatch of size M **do**
- 12: Standardise $\hat{A}$; compute $\mathcal{L}_\rho^{\text{PPO}}$ (1.25) and $\mathcal{L}_\rho^V$ (1.26).
- 13: Backpropagate, clip $\|\nabla\| \leq g_{\max}$, step $\text{Adam}_\rho^\pi, \text{Adam}_\rho^V$.
- 14: **end for**
- 15: **end for**

1.8. Proposed: FOFE-MAPPO

To make the policy agnostic to the number of visible teammates, targets and radars, the flat encoding (1.27) is replaced by a permutation-invariant set encoder, Flexible Observation Feature Encoding (FOFE), adapted from [CITE: Wang et al., 2025/2026]. The environment, MAPPO objective and reward are unchanged; only the actor and critic input pipelines are replaced.

Sequence-Exchangeable Encoding (SEE) layer. Let $X \in \mathbb{R}^{n \times d_{\text{in}}}$ be the feature matrix of a variable-size entity channel with visibility mask $m \in \{0, 1\}^n$. One SEE layer of hidden width h computes

$$\tilde{X} = \text{LayerNorm}(X), \quad (1.28)$$

$$E = \tilde{X}W_e, \quad Z = \tilde{X}W_g, \quad G = \text{SiLU}(Z), \quad (1.29)$$

$$e_{\text{agg}} = \max_{i: m_i=1} E_{i,:} \in \mathbb{R}^h, \quad (1.30)$$

$$\text{SEE}(X, m)_{i,:} = [G_{i,:} \odot E_{i,:} \parallel G_{i,:} \odot e_{\text{agg}}] \in \mathbb{R}^{2h}, \quad (1.31)$$

with layer-norm omitted in the first SEE layer. Masked rows are zeroed after concatenation. The left branch preserves per-entity local features; the right branch broadcasts a gated global summary back to every row, yielding a permutation-equivariant output.

FOFE block. A channel encoder stacks L_{SEE} SEE layers, applies a shared per-entity MLP f_ψ , and collapses the set to a fixed-size vector by a masked max-pool,

$$\text{FOFE}(X, m) = \max_{i: m_i=1} f_\psi(\text{SEE}_{L_{\text{SEE}}} \circ \dots \circ \text{SEE}_1(X, m))_i \in \mathbb{R}^{D_{\text{FOFE}}}. \quad (1.32)$$

The output is permutation-invariant and independent of n . If no entity is visible ($\sum_i m_i = 0$) the block returns a zero vector instead of propagating $-\infty$ from the masked pool.

FOFE actor. For each role, the self-observation is mapped by an MLP g_ξ , and the three entity channels are encoded by independent FOFE blocks. The resulting vectors are concatenated and passed through a fusion MLP and a factorised action head,

$$z_i = [g_\xi(o_i^{\text{self}}) \parallel \text{FOFE}_{\text{ag}} \parallel \text{FOFE}_{\text{tg}} \parallel \text{FOFE}_{\text{rd}}], \quad \ell_i = W_{\text{head}} \text{FusionMLP}(z_i). \quad (1.33)$$

The logits $\ell_i \in \mathbb{R}^{2K}$ parametrise the MultiCategorical distribution of Section 1.6.

FOFE critic. The critic receives the global state decomposed into three entity sets, *all* agents ($d=7$: position, speed, heading, ω , role flag, alive), all targets ($d=3$: position, alive), all radars ($d=4$: position, active, R^{eff}), plus a scalar episode-time feature $\tau = t/T \in [0, 1]$,

$$V_{\phi_p}(s) = W_v \text{FusionMLP}([\text{FOFE}_{\text{ag}}^c(s) \parallel \text{FOFE}_{\text{tg}}^c(s) \parallel \text{FOFE}_{\text{rd}}^c(s) \parallel \tau]). \quad (1.34)$$

Each role owns a full set of three actor FOFE blocks and three critic FOFE blocks; no weights are shared across roles. Network dimensions are listed in Table 1.7.

Table 1.7: FOFE-MAPPO network hyperparameters (per role). h_ℓ^{SEE} is the hidden width of SEE layer ℓ ; D_{FOFE} the per-entity MLP output.

Symbol	Meaning	Value
L_{SEE}	SEE layers per channel	1
h_1^{SEE}	SEE hidden width	96
D_{FOFE}	Per-entity MLP output dim	64
Per-entity MLP widths	$(h_1, \dots)$	(128, 64)
Self-MLP widths	$(h_1, \dots)$	(64, 64)
Fusion MLP widths	$(h_1, \dots)$	(256, 256)
Activation	Non-linearity	ReLU / SiLU (gates)
Logit clip	ℓ_{max}	20

Algorithm 2 FOFE-MAPPO (one training iteration).

Require: FOFE actors π_{θ_p} (1.33), FOFE critics V_{ϕ_p} (1.34), optimisers and hyperparameters as in Alg. 1.

- 1: **for** $b = 1, \dots, B$ in parallel **do**
 - 2: Reset environment b ; set $\mathbf{s}_0^{(b)} \sim p_0$.
 - 3: **for** $t = 0, \dots, T - 1$ **do**
 - 4: Compute $\mathcal{D}_t^\xi$ via (1.6)–(1.7); assemble per-channel feature tensors (X_t^ξ, m_t^ξ) .
 - 5: $\ell_i \leftarrow$ FOFE actor (1.33); sample $a_{i,t} \sim \text{MultiCat}(\ell_i)$.
 - 6: Step env; assemble global-state tensors (X^ξ, m^ξ) and τ_t ; log transition and $\log \pi_{\theta_p^{\text{old}}}$.
 - 7: **end for**
 - 8: **end for**
 - 9: For each role ρ , compute V_{ϕ_p} via (1.34) and GAE $\hat{A}_t^\rho, \hat{R}_t^\rho$ via (1.24).
 - 10: **for** $e = 1, \dots, E$ **do**
 - 11: **for** minibatch of size M **do**
 - 12: Standardise $\hat{A}$; compute $\mathcal{L}_\rho^{\text{PPO}}$ (1.25) and $\mathcal{L}_\rho^V$ (1.26); apply logit clip.
 - 13: Backpropagate, clip $\|\nabla\| \leq g_{\text{max}}$, step $\text{Adam}_\rho^\pi, \text{Adam}_\rho^V$.
 - 14: **end for**
 - 15: **end for**
-

Figure placeholder: a block diagram of the FOFE actor/critic pipeline, showing the three entity channels passing through separate FOFE blocks and concatenating with the self-embedding (and τ for the critic) before the fusion MLP, would make the architecture immediately legible.

1.9. Performance Metrics

1.9.1. Key Performance Indicators

Evaluation uses N_{eval} held-out layouts, unseen during training. The primary KPIs are

1. *Mission success rate* $\bar{s} = \frac{1}{N_{\text{eval}}} \sum \mathbb{I}[\text{all targets destroyed}]$;
2. *Survival rate per role*, $\bar{\kappa}_\rho$;
3. *Mean episode return* (per role and team);
4. *Time-to-first-kill* and *mission time* (steps until all targets are destroyed or episode timeout);

5. *Generalisation gap*: difference between $\bar{s}$ on training layouts versus held-out layouts with N_s, N_j, N_r, N_t outside the training distribution.

Figure placeholder: *a learning-curve template (mean $\pm$ s.e. over seeds, for both methods in LF and HF) and a table of final KPIs at the end of training would round out this section.*

1.9.2. Emergent Tactics

Beyond aggregate KPIs, we analyse qualitative tactical behaviour through spatial statistics of trajectories: (i) the *suppress-then-penetrate* index, defined as the fraction of successful missions in which at least one jammer enters its suppression region before the earliest striker kill; (ii) the *formation coherence* $\bar{d}^{\text{ally}}$, measured as the mean cross-role distance during the approach phase; (iii) *flanking angle*, the bearing of the striker to the radar at the moment of the kill, used to detect whether the policy learns to attack from the side-lobe sector in the HF environment.

Simulation-Based Results

2.1. Impact of Environmental Fidelity

In this section the impact of enhanced environment fidelity on emergent tactics is evaluated. Emergent tactics are evaluated using a visualisation based analysis along with quantified KPIs.

The FOFE-MAPPO method is evaluated in the low fidelity and the HF-model.

2.2. Impact of Distributed Information Sharing

In this section the impact of distributed information sharing is shared.

2.3. Ablation Study

In this section the effects of FOFE are evaluated on learning, performance and emergent tactics.

2.4. Generalization

2.5. Operational Sensitivity

3 Discussion

References

- [1] Secretary of the Air Force Public Affairs. *Air Force designates two Mission Design Series for collaborative combat aircraft*. Accessed: 2026-02-26. U.S. Air Force. Mar. 2025. URL: <https://www.af.mil/News/Article-Display/Article/4092641/air-force-designates-two-mission-design-series-for-collaborative-combat-aircraft/>.